

Magnetoelastic wave in Ferromagnetic Thin-film Mediated by Dipolar Interaction

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Abstract. The interaction of magnetic and elastic degree of freedom via magnetic dipolar interaction is investigated. The microscopic theoretical framework that deal with the dipolar interaction in ferromagnetic thin films is constructed. The derived coupled equations of motion predict mixing between magnetostatic and Lamb waves. The dispersion of the coupled wave of YIG film is calculated and shows level repulsions of MHz in size between modes with the same parities.

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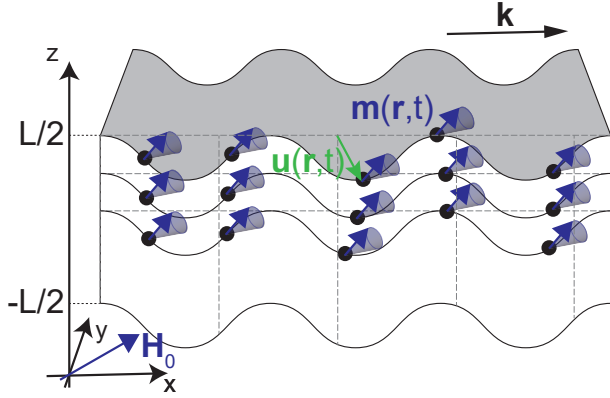


Figure 1. Schematic illustration of the system under consideration. Ferromagnetic thin-film of thickness L in the z direction and infinitely large in x and y directions. Magnetic moments at each lattice point precess around an axis parallel to the external in-plane magnetic field $\mathbf{H}_0$. The elastic wave changes positions of magnetic moments and their distance, resulting in the modification of dipolar interaction among them.

Magneto-elastic coupling is a focus of recent studies with the prospect of enhancing the manipulation of magnetic properties of materials. In quantum mechanics length scale, two degree of freedoms are connected via spin-orbit coupling, which is a local interaction and cause magnon-phonon coupling. On the other hand, coupling at classical dynamics length scale is mainly dominated by the magnetic dipolar interaction. In this case, magnetostatic waves and elastic waves are coupled. Experimental investigation of this realm is advanced thanks to the development of IDT and precise experiment is possible. However, there does not exist a microscopic theory that treats the non-local dipolar interaction thus far. In this Letter, we construct a general formalism to deal with dipolar interaction and resulting magnetostatic-elastic waves.

Our system is schematically shown in Fig. 1. Consider a ferromagnetic thin film, that is infinitely large in the xy plane and has a finite thickness L in the z direction. The origin is taken to be in the middle of the film. This film is assumed to be isotropic both magnetically and elastically. We apply an in-plane external magnetic field $\mathbf{H}_0$ to this system. Then, at equilibrium, saturation magnetization $\mathbf{M}_0$ appear parallel to $\mathbf{H}_0$. We study the propagation wave of small deviation from equilibrium magnetization

$\mathbf{m}(\mathbf{r}, t)$. This system also has another propagating wave of elastic deformation. At each position $\mathbf{r}$, the lattice points are shifted by $\mathbf{u}(\mathbf{r}, t)$. By this deformation, the distance between magnetic dipoles at each lattice point is modulated and the dipole field $\mathbf{h}(\mathbf{r}, t)$ created by $\mathbf{m}$ is also affected by $\mathbf{u}$ and the system shows the elastomagnetic wave propagation. For simplicity, we consider planewaves with frequency ω and take the x axis parallel to its propagation direction and assume that the system is uniform in the y direction. Then, $\mathbf{m}$ and $\mathbf{u}$ can be written as

$$\mathbf{m}(\mathbf{r}, t) = \mathbf{m}(z)e^{i(kx - \omega t)}, \quad \mathbf{u}(\mathbf{r}, t) = \mathbf{u}(z)e^{i(kx - \omega t)}. \quad (1)$$

We study the coupled dynamics of these waves up to the first-order of these quantities. Here, we focus exclusively on the case of in-plane magnetic field. The case of arbitrary external magnetic field will be presented elsewhere.

We start from the basic equations describing the magnetic dipolar interaction: Maxwell's equations. When the elastic deformation is present, the lattice deviates from its equilibrium positions $\mathbf{r}$ to $\mathbf{R} := \mathbf{r} + \mathbf{u}(\mathbf{r}, t)$. Then, the Maxwell's equations are rewritten in terms of this new coordinate as

$$\nabla_{\mathbf{R}} \cdot (\mathbf{H} + \mathbf{M}) = 0, \quad \nabla_{\mathbf{R}} \times \mathbf{H} \approx 0, \quad (2)$$

where in the second equation, the magnetostatic approximation is applied and the time-derivative of electric field is discarded. This approximation is justified when the length scale is long enough so that the dipolar interaction is dominant. We decompose the magnetic field $\mathbf{H}$ and the magnetization $\mathbf{M}$ as $\mathbf{H} = \mathbf{H}_0 + \mathbf{h}(\mathbf{r}, t)$ and $\mathbf{M} = \mathbf{M}_0 + \mathbf{m}(\mathbf{r}, t)$, respectively. Here, $\mathbf{H}_0$ and $\mathbf{M}_0$ are quantities at the equilibrium under in-plane external magnetic field and parallel to each other. By the planewave assumptions (1), up to the first-order, the dipolar field can also be written as $\mathbf{h} = \mathbf{h}(z)e^{i(kx - \omega t)}$. We rewrite Eq. (2) in terms of these quantities in the original coordinate $\mathbf{r}$. We note that we need to pay attention to the conservation of the magnetization in a unit volume and as a result we get $\mathbf{M}(\mathbf{R}) = \mathbf{M}(\mathbf{r}) / \det(J(\mathbf{r}))$, where $J(\mathbf{r})$ is a Jacobi matrix (see Supplemental Material [1] for detail). We get

$$\nabla_{\mathbf{r}} \cdot (\mathbf{h}(\mathbf{r}) + \mathbf{m}(\mathbf{r})) = \mathbf{M}_0 \cdot \nabla_{\mathbf{r}} (\nabla_{\mathbf{r}} \cdot \mathbf{u}(\mathbf{r})), \quad (3)$$

$$\nabla_{\mathbf{r}} \times \mathbf{h}(\mathbf{r}) = 0, \quad (4)$$

where $\nabla_{\mathbf{r}}$ stands for derivatives with respect to the coordinate $\mathbf{r}$. Because the surface of the film is also

distorted as in Fig. , boundary conditions to solve these equations are also modified by the existence of $\mathbf{u}$ as

$$h_x^{\text{in}} - h_x^{\text{out}} = M_{0,z} \frac{\partial u_z}{\partial x}, \quad (5)$$

$$h_z^{\text{in}} + m_z - h_z^{\text{out}} = M_{0,x} \frac{\partial u_z}{\partial x}, \quad (6)$$

at $z = \pm L/2$, where $h^{\text{in/out}}$ stand for the dipolar fields inside and outside of the film, respectively. Following the Green tensor method as in Ref. [2], the dipolar field satisfying Eqs. (3) and (4) under boundary conditions (5) and (6) is given as (see Supplemental Material [1] for the detail)

$$\mathbf{h}(z) = \int_{-\frac{L}{2}}^{\frac{L}{2}} dz' (G^m(z, z') \mathbf{m}(z') + G^u(z, z') \mathbf{u}(z')), \quad (7)$$

where

$$G^m(z, z') = \begin{pmatrix} -G_P(z, z') & 0 & -iG_Q(z, z') \\ 0 & 0 & 0 \\ -iG_Q(z, z') & 0 & G_{P'}(z, z') \end{pmatrix}, \quad (8)$$

$$G^u(z, z') = -ikM_{0,x}G^m(z, z'),$$

$$G_P(z, z') = \frac{k}{2} e^{-k|z-z'|}, \quad (9)$$

$$G_Q(z, z') = \text{sgn}(z - z') G_P(z, z'), \quad (10)$$

$$G_{P'}(z, z') = G_P(z, z') - \delta(z - z'). \quad (11)$$

This dipolar field couples the magnetostatic wave and elastic wave on the film. We note that in the current setup, there are no couplings when the equilibrium magnetization is parallel to the y axis, i.e. the external magnetic field is applied in that direction. Otherwise, two waves are coupled via $\mathbf{h}$.

Next, we examine the effect of the magnetoelastic coupling in the dynamics of the magnetostatic and elastic waves. In the absence of the coupling, the dynamics of magnetization $\mathbf{m}$ up to its first-order is described by the linearized Landau-Lifshitz equation

$$\dot{\mathbf{m}} = -\gamma (\mathbf{m} \times \mathbf{H}_0 + \mathbf{M}_0 \times \mathbf{h}), \quad (12)$$

where γ is the gyromagnetic ratio. This equation can be derived from a Lagrangian

$$L^m[\mathbf{m}, \dot{\mathbf{m}}, \mathbf{m}^*, \dot{\mathbf{m}}^*] = \int dz \Re \left[\frac{\mu_0}{4M_0^2 \gamma} \mathbf{M}_0 \cdot (\mathbf{m} \times \dot{\mathbf{m}}^*) - \frac{\mu_0}{4} \left(\mathbf{h}^* \cdot \dot{\mathbf{h}} + \frac{H_0}{M_0} \mathbf{m}^* \cdot \dot{\mathbf{m}} \right) \right]. \quad (13)$$

We can check it by constructing the Euler-Lagrange equation with respect to $\mathbf{m}^*$ from this Lagrangian. Here, μ_0 is the vacuum magnetic permeability, which is required to make it consistent with the energy of the magnetic field. On the other hand, the dynamics of the displacement field $\mathbf{u}$ is described by the Navier-Cauchy equation

$$\rho \ddot{\mathbf{u}} = \mu \nabla^2 \mathbf{u} + (\mu + \lambda) \nabla (\nabla \cdot \mathbf{u}), \quad (14)$$

where ρ is the mass density, μ and λ are Lamé constants. The Lagrangian for this equation is

$$L^u[\mathbf{u}, \dot{\mathbf{u}}, \mathbf{u}^*, \dot{\mathbf{u}}^*] = \int dz \left[\frac{\rho}{4} \dot{\mathbf{u}}^* \cdot \dot{\mathbf{u}} + \frac{1}{4} \partial_i \sigma_{ij} u_j^* \right], \quad (15)$$

where $\sigma_{ij} = \mu(\partial_i u_j + \partial_j u_i) + \lambda \delta_{ij} \nabla \cdot \mathbf{u}$ is the stress tensor. The total Lagrangian of a coupled system is given by the sum of two Lagrangians (13) and (15). In this case, $\mathbf{h}$ is no longer just a functional of the amagnetization, but also a functional of the displacement field $\mathbf{u}$. Therefore, there appear a new volume force term $-\mu_0 \mathbf{f}$ in the Navier-Cauchy equation (14), where the force $\mathbf{f}$ is given as (see Supplemental Material [1] for the detail of derivation)

$$f_i(z) := \frac{\delta h_j^*}{\delta u_i^*} h_j = -(\mathbf{M}_0 \cdot \nabla) h_i(z). \quad (16)$$

Physically, this force can be understood as a Maxwell stress acting on magnetic dipoles at each lattice site caused by the change of the dipolar field, which is induced by the displacement field. The coupled dynamics of the magnetoelastic wave is then given by

$$\begin{cases} \dot{\mathbf{m}} = -\gamma (\mathbf{m} \times \mathbf{H}_0 + \mathbf{M}_0 \times \mathbf{h}) \\ \rho \ddot{\mathbf{u}} = \mu \nabla^2 \mathbf{u} + (\mu + \lambda) \nabla (\nabla \cdot \mathbf{u}) - \mu_0 \mathbf{f}. \end{cases} \quad (17)$$

In order to see the dynamical property of the coupled wave described by the above equations of motion, we numerically calculate the dispersion relation below. By the assumption (1), the time derivative in the left-hand side becomes $-i\omega$. The second equation of Eq. (17) contains spatial derivative of $\mathbf{u}$ and requires the boundary condition to solve this equation. Since we consider a film in the vacuum, we impose the tension free condition at the surfaces $z = \pm L/2$ as

$$\sigma_{xz}|_{z=\pm \frac{L}{2}} = \sigma_{yz}|_{z=\pm \frac{L}{2}} = \sigma_{zz}|_{z=\pm \frac{L}{2}} = 0. \quad (18)$$

Without coupling, it is known that a type of elastic wave called Lamb wave appear under these conditions. Since the system is symmetric with respect to the $z = 0$ plane, modes of the Lamb wave are either symmetric or anti-symmetric with respect to the sign change of the variable z . Numerically, we need to solve the equation while satisfying the boundary conditions (18). It is well known that the finite element method provides naturally incorporate these conditions, for example. On the other hand, the magnetostatic wave without coupling in this geometry is studied in detail by Damon and Eshbach [3]. They analytically showed that when the external magnetic field is parallel to the propagation of the wave ($\mathbf{H}_0 \parallel \mathbf{k}$), the system has infinitely many bulk mode called backward volume wave (BVW), which are either symmetric or anti-symmetric. When the magnetic field is in-plane but perpendicular to the propagation ($\mathbf{H}_0 \perp \mathbf{k}$), there will be a surface mode, which is localized at the surface. For the calculation of the dispersion, it is more convenient

to introduce another coordinate (r_1, r_2, r_3) , where the original (x, y, z) is rotated around the z axis and r_1 axis is taken to be parallel to the external magnetic field. Then, r_1 component of $\mathbf{m}$ is zero since $\mathbf{M}_0 \cdot \dot{\mathbf{m}} = 0$ in the first equation of Eq. (17) and the equation is reduced to an equation about two-component vector $(m_2, m_3)^T$.

Now we consider the coupled equations (17). Using the finite element method, we discretize the equation in real space while satisfying the boundary conditions, we get a generalized eigenvalue problem about the eigenvalue ω . By solving this equation, get dispersion curves as plotted in Fig. 2. The parameters are taken to be that of Yttrium-Iron-Garnet (YIG), $\rho = 5.17 \text{ g/cm}^3$, $\mu = 7.81 \times 10^{-8} \text{ kg} \cdot \text{GHz}^2/\text{m}$, $\lambda = 10.94 \times 10^{-8} \text{ kg} \cdot \text{GHz}^2/\text{m}$ [4], $\gamma = 29.8 \text{ GHz/T}$, $\mu_0 M_0 = 0.178 \text{ T}$ [5] and the vacuum permeability is $\mu_0 = 4\pi \times 10^{-11} \text{ T}^2\text{m}/(\text{kg} \cdot \text{GHz}^2)$. We set the thickness of the system to be $L = 500 \text{ }\mu\text{m}$ and the size of the external magnetic field to be $\mu_0 H_0 = 0.1 \text{ T}$. We define the angle between the x axis and the external magnetic field as φ and write $\mathbf{H}_0 = H_0(\cos \varphi, \sin \varphi, 0)$. In Fig. 2 (a), we plot the case of $\varphi = 0$. The magnetostatic BVWs appear in this case, which are plotted by blue and red curves, which represent anti-symmetric and symmetric modes, respectively. Similarly, anti-symmetric and symmetric Lamb waves with positive group velocities (positive slopes) are plotted by red and blue, respectively. The dots show the dispersion of the coupled wave. The insets show the magnified images of the bands at crossings of BVW and Lamb waves. We can observe that the magnetostatic and elastic waves couple and make level repulsions of the size of several MHz only when two modes have the same parities. Fig. 2 (b) shows the case of $\varphi = \pi/4$. In this case, magnetostatic wave has two kinds of coexisting modes, BVW and surface wave modes. The insets show the magnified images of the band crossing points of magnetostatic surface mode and Lamb wave modes. In this case, the magnetostatic surface mode is localized at the surface of the system and it is neither anti-symmetric nor symmetric. Thus, the Lamb modes couple with this mode irrespective to their parities. Finally, in Fig. 2 (c), we plot the case of $\varphi = 0$. In this case, magnetostatic surface mode appear but it does not couple with the Lamb modes. This is because $G^u = 0$, $\mathbf{f} = \mathbf{0}$ and Eq. (17) becomes just two independent equations.

In conclusion, we constructed a framework to treat the magnetoelastic coupling mediated by the magnetic dipolar interaction in ferromagnetic thin films. The dipolar field is derived from the Maxwell's equation with distortion in lattice, which modifies the form of the equation from that of the system without distortion. Then, coupled equations of motion are

derived using Lagrangian formalism. These equations are solved numerically assuming the thin film made of YIG. We get the dispersion curves of the coupled wave. For magnetostatic bulk backward modes, modes with the same parity as the elastic wave couples and produce gaps of the size of several MHz. For magnetostatic surface mode, it couples with elastic wave regardless of the parity and also produce level repulsion of 0.1 to several MHz. The formulation constructed here can be a standing point for theoretical study of dipole interaction mediated magnetoelastic couplings.

Acknowledgments

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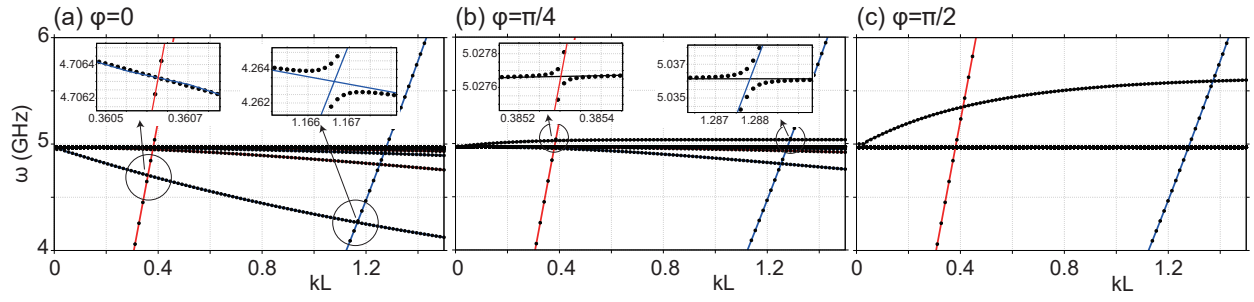


Figure 2. Dispersion curves of the coupled wave. Red and blue curves show symmetric and anti-symmetric modes of uncoupled waves, respectively. The black dots represent the dispersion of the coupled wave. The direction of the external magnetic field is (a) $\mathbf{H}_0 \parallel \mathbf{k}$ ($\varphi = 0$), (b) $\varphi = \pi/4$, and (c) $\mathbf{H}_0 \perp \mathbf{k}$ ($\varphi = \pi/2$). The insets are magnifications of some of the band crossings. Level repulsion occurs at crossings of same colored curves i.e. between modes with the same parity.